Equilibrium-Whitened Gaussian Natural Gradient Flow: A Certified Bracket, a Correction to the Lipschitz Frontier, and the Remaining Excess-Coupling Lemma

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Abstract

We report the outcome of concurrent prove and refute searches, followed by bidirectional adversarial audit. The proposed Lipschitz formula is *false as a global statement*: already in dimension one,

$$\gamma^{-1} \le 3 + C \min\{\log \kappa, \log(e + L_H), L_H^2\},$$

so an unconditional additive $\log \kappa$ is impossible at fixed L_H. An explicit smooth scalar family has arbitrary prescribed optimal condition number and prescribed fixed L_H, so this is not a vacuous parameter issue. Moreover, the two proposed formulas are mutually inconsistent without a dimension cap, because the Lipschitz class is a subclass of the bounded Hessian class.

Neither the dimension-only frontier nor the corrected higher-dimensional Lipschitz frontier is closed here. The best certified parameter-only upper bounds obtained in this run are

$$\gamma^{-1} \le C \left[1 + \min\left\{\sqrt{\kappa}, \sqrt{N \log(e\kappa)} + N^{1/3} \log(e\kappa)\right\}\right]$$

under (H1), and the stronger multi-branch estimate in Theorem 3.6 under (H1)+(H2). The latter includes the new rank-adaptive bound

$$\tau_H^2 \le \min\{(1-\alpha)(\beta-1), (1-\alpha)(1-\alpha+\sqrt{N}L_H), (\beta-1)(\beta-1+\sqrt{N}L_H)\}.$$

On the lower side, a floor-lowered version of the exact two-scale radial-softplus family has, uniformly for $0 < \alpha \le \sigma/R$,

$$N \asymp R^2, \quad \kappa \asymp \frac{R}{\alpha\sigma}, \quad L_H \asymp \frac{R}{\sigma^2}, \quad \tau_H^2 \asymp \frac{R}{\sigma}, \quad \gamma^{-1} \asymp \frac{R}{\sigma} = N^{1/4} L_H^{1/2}.$$

It therefore certifies the dimension/Lipschitz face even for $\kappa > N$. The report ends with proved rigidity lemmas, quantitative obstruction profiles, and one standalone excess-coupling lemma whose resolution would close the corrected gap.

Contents

1 Verdict and proof-status map

Let $Z \sim N(0, I_N)$, $W = \nabla^2 V$, and assume

$$\alpha I \preceq W \preceq \beta I, \qquad \mathbb{E}W = I, \qquad \kappa = \beta/\alpha. \tag{1.1}$$

Thus $0 < \alpha \le 1 \le \beta$. Under (H1), W is understood as the bounded weak Hessian of V, defined almost everywhere; under (H2), it has its operator-Lipschitz representative. When imposed, (H2) means

$$\|W(x) - W(y)\|_{\mathrm{op}} \le L_H \|x - y\|. \tag{1.2}$$

All constants below are numerical. Curvature constants and Lipschitz moduli are optimal unless explicitly described as upper bounds.

Define the worst inverse gaps

$$\mathfrak{R}_D(N,\kappa) := \sup_{\text{(H1)}} \gamma^{-1}, \qquad \mathfrak{R}_L(N,\kappa,L) := \sup_{\substack{\text{(H1),(H2)} \\ L_H \le L}} \gamma^{-1}.$$

The inclusions alone give

$$\mathfrak{R}_L(N,\kappa,L) \le \mathfrak{R}_D(N,\kappa). \tag{1.3}$$

Thus the two formulas in the question cannot both be globally sharp: when $N < \kappa$ and L is large, the proposed Lipschitz expression contains $\sqrt{\kappa}$, whereas the proposed dimension-only expression contains $\sqrt{N}$. This observation does *not* prove a $\sqrt{N}$ upper bound; that is part of the open dimension-only problem.

There is, however, an unconditional refutation.

Theorem 1.1 (The original F–L formula is false)**.** *For $N = 1$, every admissible W satisfying (H1)+(H2) obeys*

$$\gamma^{-1} \le 3 + C \min\{\log\kappa, \log(e + L_H), L_H^2\}. \tag{1.4}$$

For every $\kappa > 1$ and $L_0 > 0$, the class with optimal condition number exactly κ and optimal modulus exactly L_0 is nonempty. Hence, at fixed L_0, its worst inverse gap is uniformly bounded as $\kappa \to \infty$, contradicting the proposed unavoidable $\log\kappa$.

The proof is in Section 4. The evidence and all certified families are consistent with the following *corrected conjecture*:

$$\mathfrak{R}_D(N,\kappa) \asymp 1 + \log\kappa + \sqrt{\min\{N,\kappa\}}, \tag{1.5}$$

$$\mathfrak{R}_L(N,\kappa,L) \asymp 1 + \min\{\log\kappa, \log(e+L)\} + \min\{\sqrt{\kappa}, \sqrt{N}, N^{1/4}L^{1/2}\}. \tag{1.6}$$

Here and below additive positive terms may equivalently be replaced by their maximum, up to a factor of three. Formula (1.6), including its $\sqrt{N}$ cap, remains conjectural and depends on closing the dimension-only problem. No claim in this report assumes it.

Certified ledger

The shared ledger contains only items that survived cross-audit. Later sections cite these labels.

Label	Certified item
C1	Exact Bochner block, packed Rayleigh identity, coupling gap $\gamma \geq (3+\tau_H^2)^{-1}$, and curvature floor $\gamma \geq \alpha$.
C2	Two-sided shifted Schur bound $\tau_H^2 \leq (1-\alpha)(\beta-1)$.
C3	Under (H2), for unit w, the rank-adaptive shifted estimates $S_w \leq a(a+L_H\rho_w)$ and $S_w \leq b(b+L_H\rho_w)$.
C4	Scalar directional cap $\\|M_w\\|_{\mathrm{op}}^2 \leq C\min\{\log\kappa, \log(e+L_H), L_H^2\}$.
C5	Under (H1), entropy self-consistency: $\tau_H^2 \leq C[1+\sqrt{N\log\beta}+N^{1/3}\log\beta]$, with shifted versions and an (H2)-refined operator-scale term.
C6	The parameter-only upper brackets in Theorem 3.6.
C7	The exactly isotropized two-scale radial-softplus family and its affine Rayleigh and first-Hermite certificates.
C8	Floor interpolation: $L_\star(W_a) = aI + (1-a)L_\star(B)$, with exact scaling of γ, τ_H, L_H.
R1	Convex mixtures, rotations, direct sums, and tensor products do not amplify a bad gap.
R2	Shared-longitudinal modules obey the exact isotropy budget $\sum_j a_j/m_j = 1$; identical star replication gains no power.
R3	Common-slope multi-ramp radial families collapse to one tilted barycenter and obey an amplitude/entropy cap.
R4	Positive fixed-frame ridge banks obey a POVM/Kadison coupling bound and an exact conditional affine-energy identity.
R5	A globally bounded Hessian cannot contain a nonzero unbounded flat quadratic transverse plateau; the clipped version pays an inverse-radius amplitude cost.
O1–O5	Quantitative obstruction profiles in Section 7.

2 Exact operator identities

Let Sym_N carry the Frobenius inner product. Define

$$T[X] := \mathbb{E}[\mathrm{Tr}(XW(Z))Z], \qquad T^*[u] := \mathbb{E}[(u^\top Z)W(Z)], \tag{2.1}$$

$$H[X] := \mathbb{E}[W(Z)(Z^\top XZ - \mathrm{Tr}\,X)]. \tag{2.2}$$

After packing the physical covariance perturbation as $X/\sqrt{2}$,

$$L_\star = \begin{pmatrix} I & T/\sqrt{2} \\ T^*/\sqrt{2} & I + H/2 \end{pmatrix}, \qquad \gamma = \lambda_{\min}(L_\star), \qquad \tau_H = \|T\|_{\mathrm{op}}. \tag{2.3}$$

Write

$$M_w := T^* w = \mathbb{E}[(w^\top Z)W(Z)], \qquad S_w := \|M_w\|_F^2. \tag{2.4}$$

Lemma 2.1 (Bochner and Rayleigh identities; C1)**.** *For a genuine Hessian satisfying $\mathbb{E}W = I$,*

$$\mathbb{E}[(u + XZ)^\top W(Z)(u + XZ)] = \|u\|^2 + 2u^\top T[X] + \|X\|_F^2 + \langle X, H[X]\rangle_F. \tag{2.5}$$

Consequently

$$\begin{pmatrix} I & T \\ T^* & I + H \end{pmatrix} \succeq 0, \qquad I + H - T^*T \succeq 0. \tag{2.6}$$

More generally, if a positive semidefinite genuine Hessian field G has $\mathbb{E}G = cI$, and T_G, H_G are defined as in (2.1)–(2.2), then

$$\mathbb{E}[(u + XZ)^\top G(Z)(u + XZ)] = c\|u\|^2 + 2u^\top T_G[X] + c\|X\|_F^2 + \langle X, H_G[X]\rangle_F, \tag{2.7}$$

$$\begin{pmatrix} cI & T_G \\ T_G^* & cI + H_G \end{pmatrix} \succeq 0. \tag{2.8}$$

In physical covariance coordinates,

$$4\mathcal{Q}_W(u, X) = \mathbb{E}[(2u + XZ)^\top W(Z)(2u + XZ)] + \|X\|_F^2. \tag{2.9}$$

In particular,

$$\gamma \geq \alpha, \qquad \gamma \geq \frac{1}{3 + \tau_H^2}. \tag{2.10}$$

Proof. For smooth V, one and two Gaussian integrations by parts, followed by third- and fourth-derivative symmetry, give

$$\mathbb{E}[u^\top W XZ] = u^\top T[X], \qquad \mathbb{E}[(XZ)^\top W(XZ)] = \|X\|_F^2 + \langle X, H[X]\rangle_F.$$

The same calculation with $\mathbb{E}G = cI$ replaces each identity contribution by c and proves (2.7). This proves (2.5), (2.7), and (2.9). Positivity of the corresponding left sides gives the two blocks and the Schur complement in (2.6). The bound $W \succeq \alpha I$ in (2.9) gives the α-floor.

For the coupling bound, (2.6) implies

$$I + H/2 \succeq (I + T^*T)/2.$$

On a singular-value sector of T, the resulting lower comparison operator is

$$\begin{pmatrix} 1 & s/\sqrt{2} \\ s/\sqrt{2} & (1 + s^2)/2 \end{pmatrix}.$$

It has determinant $1/2$ and trace $(3 + s^2)/2$, so its smaller eigenvalue is at least $(3 + s^2)^{-1}$. Take $s \leq \tau_H$.

Under (H1) without classical derivatives, apply the stationary OU mollification $W_s = P_s W = \nabla^2(e^{2s} P_s V)$. It preserves the curvature bounds and $\mathbb{E}W_s = I$. The identities hold for $s > 0$ and pass to the limit by dominated convergence, using bounded W times a Gaussian-polynomial dominator. This is the only regularization used in the (H1)-only arguments. ∎

Lemma 2.2 (Shifted Schur cancellation; C2)**.** *Put*

$$a = 1 - \alpha, \qquad b = \beta - 1. \tag{2.11}$$

Then

$$\tau_H^2 \leq ab = (1 - \alpha)(\beta - 1). \tag{2.12}$$

Moreover

$$\gamma^{-1} \leq 3 + \sqrt{\kappa}. \tag{2.13}$$

Proof. Apply the generalized Bochner block (2.8) to $A = W - \alpha I \succeq 0$ and to $C = \beta I - W \succeq 0$. Their means are aI, bI, their couplings are $T, -T$, and their fourth-order terms are $H, -H$. For $a, b > 0$, their Schur complements give

$$T^*T \preceq a(aI + H), \qquad T^*T \preceq b(bI - H).$$

If either scalar is zero, positivity of the corresponding block forces $T = 0$, so the same inequalities hold at the endpoint. If $a + b = 0$, then $W = I$ almost surely and the conclusion is immediate. Otherwise, the convex combination with weights $b/(a + b)$ and $a/(a + b)$ cancels H and gives $T^*T \preceq abI$. Notice that (2.12) does *not* say $\tau_H^2 \le \sqrt{\kappa}$.

If $\alpha \ge \kappa^{-1/2}$, the floor gives $\gamma^{-1} \le \sqrt{\kappa}$. Otherwise $\beta = \kappa\alpha < \sqrt{\kappa}$, so $ab \le \beta < \sqrt{\kappa}$ and the coupling branch gives $\gamma^{-1} \le 3 + \sqrt{\kappa}$. □

3 Certified upper side

This section proves the strongest parameter-only upper envelope found in the run. It does not close either proposed frontier.

3.1 Where (H2) enters: a weak third-derivative estimate

Under (H2), Rademacher's theorem gives $\nabla^3 V$ almost everywhere and

$$\|\nabla^3 V(x)\|_{\mathrm{inj}} \le L_H \quad \text{for a.e. } x. \tag{3.1}$$

No pointwise fourth derivative is assumed in the next lemma; the needed fourth-order compatibility identity is used only in the weak/OU-mollified form of Lemma 2.1.

Lemma 3.1 (Rank-adaptive Lipschitz–Schur estimate; C3)**.** *For a unit vector w with $S_w > 0$, let*

$$B_w = M_w/\sqrt{S_w}, \qquad \rho_w = \|B_w\|_* \le \sqrt{\operatorname{rank} M_w} \le \sqrt{N}.$$

Then

$$S_w \le a(a + L_H\rho_w), \qquad S_w \le b(b + L_H\rho_w). \tag{3.2}$$

Consequently

$$\tau_H^2 \le \min\{ab, a(a + \sqrt{N}L_H), b(b + \sqrt{N}L_H)\}. \tag{3.3}$$

Proof. For $q_B(Z) = Z^\top BZ - \operatorname{Tr} B$, Hessian compatibility and the generalized Bochner identity (2.7) give

$$Q_A(B) := \mathbb{E}[(BZ)^\top A(BZ)] = a\|B\|_F^2 + \mathbb{E}[q_B \operatorname{Tr}(BA)]. \tag{3.4}$$

The shifted Schur complement yields

$$S_w = \langle B_w, T^*w\rangle_F^2 \le \|TB_w\|^2 \le aQ_A(B_w).$$

Weak Gaussian integration by parts gives

$$\mathbb{E}[q_B \operatorname{Tr}(BA)] = \mathbb{E}[(BZ)^\top \nabla \operatorname{Tr}(BA)]. \tag{3.5}$$

Diagonalizing B and using (3.1),

$$\|\nabla \operatorname{Tr}(BW)\| \le L_H\|B\|_*.$$

Since $\mathbb{E}\|BZ\| \le \|B\|_F = 1$, equations (3.4)–(3.5) prove the first bound. Applying the same argument to $C = \beta I - W$ proves the second. This is the precise point where (H2) enters. □

For unit w, direct first-order Stein also gives

$$M_w = \mathbb{E}[D_w W(Z)], \qquad \|M_w\|_{\mathrm{op}} \le L_H, \qquad S_w \le N L_H^2. \tag{3.6}$$

Combining (3.3) with C1–C2 already yields

$$\gamma^{-1} \le 4 + \min\{\sqrt{\kappa}, \sqrt{N} L_H, N L_H^2\}. \tag{3.7}$$

The first Lipschitz branch is power-slack on the mandatory two-scale audit; Section 8 records the exact loss.

3.2 Entropy self-consistency under (H1)

Lemma 3.2 (Bounded-density quadratic tilt). *Let $p \ge 0$, $\mathbb{E}p = 1$, $p \le \beta$, and $h = \log \beta$. If $B \in \mathrm{Sym}_N$, $\|B\|_F = 1$, and $\eta = \|B\|_{\mathrm{op}}$, then*

$$|\mathbb{E}[p q_B]| \le C(\sqrt{h} + \eta h). \tag{3.8}$$

For a unit vector w,

$$|\mathbb{E}[p(w^\top Z)]|^2 \le 2h. \tag{3.9}$$

Proof. Because $p \le \beta$, its Gaussian relative entropy satisfies $D(p\gamma\|\gamma) = \mathbb{E}[p \log p] \le h$. If λ_i are the eigenvalues of B, then

$$\log \mathbb{E} e^{\theta q_B} = \sum_i \left[-\theta \lambda_i - \frac{1}{2} \log(1 - 2\theta\lambda_i) \right] \le \frac{C\theta^2}{1 - 2|\theta|\eta}$$

whenever $2|\theta|\eta < 1$. Entropy variational duality, applied to q_B and $-q_B$, followed by $\theta \asymp \min\{\sqrt{h}, \eta^{-1}\}$, proves (3.8). For the linear variable, use $\log \mathbb{E} e^{\theta w^\top Z} = \theta^2/2$ and optimize exactly. $\square$

Theorem 3.3 (Dimension/entropy bound; C5). *Under (H1),*

$$\tau_H^2 \le C \left[1 + \sqrt{N \log \beta} + N^{1/3} \log \beta \right]. \tag{3.10}$$

More precisely, when $a, b > 0$, set

$$h_a = \log \frac{\beta - \alpha}{1 - \alpha}, \qquad h_b = \log \frac{\beta - \alpha}{\beta - 1}, \tag{3.11}$$

one also has

$$\tau_H^2 \le C a^2 [1 + \sqrt{N h_a} + N^{1/3} h_a], \tag{3.12}$$

$$\tau_H^2 \le C b^2 [1 + \sqrt{N h_b} + N^{1/3} h_b]. \tag{3.13}$$

If $a = 0$ or $b = 0$, the bounds and isotropic mean force $W = I$ almost surely, so $\tau_H = 0$ and the endpoint claim is trivial.

Proof. Fix a unit vector w and write $S = S_w > 0$, $B = M_w/\sqrt{S}$. By C1,

$$S \le \|TB\|^2 \le Q_W(B) = 1 + \mathbb{E}[q_B \operatorname{Tr}(BW)]. \tag{3.14}$$

Decompose $B = B_+ - B_-$ and, whenever $\operatorname{Tr} B_\pm > 0$, set

$$p_\pm = \frac{\operatorname{Tr}(B_\pm W)}{\operatorname{Tr} B_\pm}.$$

These are densities with $p_\pm \le \beta$. Lemma 3.2 and $\|B\|_* \le \sqrt{N}$ therefore imply

$$S \le 1 + C\sqrt{N}(\sqrt{h} + \eta h), \qquad h = \log\beta, \quad \eta = \|B\|_{\mathrm{op}}. \tag{3.15}$$

For every unit v, the scalar function $p_v = v^\top W v$ is also a density bounded by β. Equation (3.9) gives

$$\|M_w\|_{\mathrm{op}}^2 \le 2h, \qquad \eta \le \sqrt{2h/S}. \tag{3.16}$$

Thus

$$S \le 1 + C\sqrt{Nh} + C\frac{\sqrt{N}h^{3/2}}{\sqrt{S}}. \tag{3.17}$$

If $S \le 2(1 + C\sqrt{Nh})$ we are done. Otherwise the last term is at least $S/2$, whence $S^{3/2} \le C\sqrt{N}h^{3/2}$ and $S \le CN^{1/3}h$. This proves (3.10).

Apply the same proof to $(W - \alpha I)/a$ and to $(\beta I - W)/b$, using the shifted Schur inequalities before normalization, to obtain (3.12)–(3.13). The OU limiting argument in Lemma 2.1 justifies all compatibility identities under (H1) alone. □

3.3 Scalar directional cap and the combined theorem

Lemma 3.4 (Nonnegative Lipschitz density)**.** *If $f \ge 0$, $m = \mathbb{E}f(Z) > 0$, and f is L-Lipschitz on $\mathbb{R}$, then*

$$\left|\frac{\mathbb{E}[Zf(Z)]}{m}\right|^2 \le C\log\left(e + \frac{L}{m}\right). \tag{3.18}$$

Proof. Positivity and Lipschitzness imply $f(0) \le C(m + L)$, by integrating the tent lower bound for f near zero. Hence $f(x) \le C[m + L(1 + |x|)]$. Under the probability density $f\varphi/m$, split the first absolute moment at $R = 2\sqrt{\log(e + L/m)}$. The bulk is at most R; the displayed envelope and the standard Gaussian tail moments bound the tail by a numerical constant. The absolute mean is no larger than this first moment. □

Lemma 3.5 (Directional operator cap; C4)**.** *Under (H1)+(H2), for every unit w,*

$$\|M_w\|_{\mathrm{op}}^2 \le C\min\{\log\kappa, \log(e + L_H), L_H^2\}. \tag{3.19}$$

Proof. Fix a unit v, let $G = w^\top Z$, and condition $v^\top W(Z)v$ on $G = t$. The resulting scalar function $f(t)$ is nonnegative, has mean one, is bounded by $\beta \le \kappa$, and is L_H-Lipschitz. The entropy-linear estimate gives $|\mathbb{E}[Gf(G)]|^2 \le 2\log\beta$; Lemma 3.4 gives the $\log(e + L_H)$ bound; and one-dimensional Stein gives the L_H^2 bound. Taking the supremum over v proves the claim. □

The last lemma gives both

$$S_w \le CN\min\{\log\kappa, \log(e + L_H), L_H^2\} \tag{3.20}$$

and a refinement of the cubic proof. Namely, replace the numerator $2h$ in (3.16) by the directional cap while retaining the quadratic-density entropy h. This proves the following summary.

Theorem 3.6 (Best certified parameter-only upper bracket; C6)**.** *Set*

$$h_\kappa = \log(e\kappa), \qquad d_{\kappa,L} = \min\{\log\kappa, \log(e + L), L^2\}. \tag{3.21}$$

Under (H1),

$$\boxed{\gamma^{-1} \le C\left[1 + \min\left\{\sqrt{\kappa}, \sqrt{Nh_\kappa} + N^{1/3}h_\kappa\right\}\right].} \tag{3.22}$$

Under (H1)+(H2),

$$\boxed{\begin{aligned}\gamma^{-1} \le C\Big[1 + \min\{\sqrt{\kappa}, \sqrt{N}L_H, NL_H^2, Nd_{\kappa,L_H},\\ \sqrt{Nh_\kappa} + N^{1/3}h_\kappa^{2/3}d_{\kappa,L_H}^{1/3}\}\Big].\end{aligned}} \tag{3.23}$$

The sharper a, b, β-dependent statements are (2.12), (3.2), and (3.12)–(3.13).

Proof. Use C1 to convert each upper bound on τ_H^2 into an upper bound on γ^{-1}, and combine with the α-floor case split in C2. Equation (3.22) follows from Theorem 3.3 and $\beta \le \kappa$. For (3.23), take the minimum of (3.7), (3.6), (3.20), and the refined cubic inequality

$$S_w \le C\left[1 + \sqrt{Nh_\kappa} + N^{1/3}h_\kappa^{2/3}d_{\kappa,L_H}^{1/3}\right].$$

All constants are absorbed by the outer $C[1 + \cdot]$. □

4 The scalar correction is exact in its parameters

For $N = 1$, $M_1 = \mathbb{E}[ZW]$ is scalar, so Lemma 3.5 and C1 immediately prove (1.4). We now certify nonvacuity for arbitrary exact (κ, L_H).

Proposition 4.1 (Exact-κ, exact-L_H scalar family)**.** *Fix $\kappa > 1$ and $L_0 > 0$. Put*

$$\alpha = \frac{2}{\kappa+1}, \qquad \varepsilon = \frac{1-\alpha}{2L_0}, \qquad B_\varepsilon(x) = \frac{2}{1+e^{x/\varepsilon}}, \qquad W_\kappa = \alpha + (1-\alpha)B_\varepsilon. \tag{4.1}$$

Then W_κ is a smooth genuine one-dimensional Hessian with

$$\mathbb{E}W_\kappa = 1, \quad \inf W_\kappa = \alpha, \quad \sup W_\kappa = 2-\alpha = \kappa\alpha, \quad \mathrm{Lip}(W_\kappa) = L_0. \tag{4.2}$$

After an affine correction, its potential satisfies $\mathbb{E}V'(Z) = 0$ and is normalizable. Its gap is bounded below uniformly in κ for fixed L_0.

Proof. The identity $B_\varepsilon(x) + B_\varepsilon(-x) = 2$ and Gaussian symmetry give $\mathbb{E}B_\varepsilon = 1$. Its range is $(0, 2)$ and its optimal derivative bound is $(2\varepsilon)^{-1}$. Equations (4.2) follow directly. Integrate twice to obtain $V'' = W_\kappa$ and choose the affine term to center $\mathbb{E}V'(Z)$. Since $W_\kappa \ge \alpha > 0$, the potential is strongly convex and normalizable. The uniform gap lower bound is also a direct consequence of (1.4). This completes the proof of Theorem 1.1. □

The compact-tail/remote-cap scalar construction from the supplied counterexample note gives the matching adverse mechanism

$$\gamma^{-1} \gtrsim 1 + \min\{\log\kappa, \log(e + L_H)\} \tag{4.3}$$

up to numerical rescaling of the logarithms: a principal ramp of height A creates a tilted Gaussian mean of order $\sqrt{\log A}$, while a very remote ramp fixes the exact supremum and infimum with negligible first two Gaussian moments. Taking A comparable to the smaller permitted height and slope produces (4.3). Lemma 3.4 is the matching obstruction. Thus the logarithmic part of the corrected conjecture is already sharp in one dimension.

5 Best certified adverse family

We give a self-contained certificate for the two-scale family and then lower its curvature floor. Put $z = (t, y) \in \mathbb{R} \times \mathbb{R}^d$, assume $\sigma \ge 1$ and $d \ge C_0\sigma^2$ for a sufficiently large numerical C_0, and, for $0 \le \delta \le 1$, define

$$r(y) = \sqrt{\sigma^2 + \|y\|^2}, \quad R = \sqrt{\sigma^2 + d}, \quad F_\sigma(s) = \sigma \log(1 + e^{s/\sigma}), \quad U_\delta(t, y) = F_\sigma(r(y) - R - \delta t). \tag{5.1}$$

For expectations below, write $Z = (T, Y)$ with $T \sim N(0, 1)$ and $Y \sim N(0, I_d)$ independent. Write $p = F'_\sigma$, $q = F''_\sigma$, $s = r - R - \delta t$, and $A_\delta = \nabla^2 U_\delta$. With

$$v = (-\delta, y/r), \qquad D = I_d/r - yy^\top/r^3,$$

the exact Hessian is

$$A_\delta = q(s)vv^\top + p(s)\begin{pmatrix} 0 & 0 \\ 0 & D \end{pmatrix} \succeq 0, \qquad 0 \preceq A_\delta \preceq \frac{3}{2\sigma}I. \tag{5.2}$$

For $S = \|Y\|^2$, define

$$Q(\delta) = \mathbb{E}q(s), \tag{5.3}$$

$$M(\delta) = \frac{1}{d}\mathbb{E}\left[q(s)\frac{S}{r^2} + p(s)\left(\frac{d-1}{r} + \frac{\sigma^2}{r^3}\right)\right]. \tag{5.4}$$

Rotation invariance gives

$$\mathbb{E}A_\delta = \operatorname{diag}(\delta^2 Q(\delta), M(\delta)I_d). \tag{5.5}$$

Lemma 5.1 (Exact isotropizing root; C7)**.** *There is a root $\delta \in (0, 1)$ of*

$$\delta^2 Q(\delta) = M(\delta) =: m, \tag{5.6}$$

and every such root satisfies

$$Q \asymp \sigma^{-1}, \qquad M \asymp R^{-1}, \qquad \delta^2 \asymp \frac{\sigma}{R}, \qquad m \asymp \frac{1}{R}, \qquad \mathbb{E}A = mI_{d+1}. \tag{5.7}$$

Moreover

$$\sup_z \|\nabla^3 U(z)\|_{\text{inj}} \asymp \sigma^{-2}. \tag{5.8}$$

Proof. On the event $|S - d| \le 2\sqrt{d}$, $|T| \le 1$, which has probability bounded below, one has $|r - R| \le 2$ and $|s| \le 3$. Hence $q \gtrsim \sigma^{-1}$ and $p \gtrsim 1$, proving the lower bounds for Q and M. The upper bounds use $q \le (4\sigma)^{-1}$ and

$$\mathbb{E}r^{-1} \le (\mathbb{E}S^{-1})^{1/2} = (d-2)^{-1/2} \lesssim R^{-1}.$$

The function $\delta^2 Q - M$ is continuous, negative at zero, and positive at one once R/σ is a sufficiently large numerical constant. The intermediate value theorem and the scalar bounds give (5.7).

Finally,

$$\|\nabla s\| \le \sqrt{2}, \qquad \|\nabla^2 s\|_{\text{op}} \le \sigma^{-1}, \qquad \|\nabla^3 s\|_{\text{inj}} \le 6\sigma^{-2},$$

while $|F'''_\sigma| \lesssim \sigma^{-2}$ and $F''_\sigma \lesssim \sigma^{-1}$. The third-order chain rule gives the upper bound in (5.8). For the reverse bound, take $y = (\sigma/2)e_1$ and send $t \to -\infty$; the softplus derivative tends to one and $\nabla^3 U \to \nabla^3 r$, whose (e_1, e_1, e_1) entry has magnitude c/σ^2. $\square$

Fix the root and set

$$B = A/m, \qquad \mathbb{E}B = I. \tag{5.9}$$

For an arbitrary floor $0 < \alpha < 1$, define

$$W_\alpha = \alpha I + (1-\alpha)B, \qquad V_\alpha(z) = \frac{\alpha}{2}\|z\|^2 + \frac{1-\alpha}{m}U(z) + b_\alpha^\top z, \tag{5.10}$$

where $b_\alpha = -(1-\alpha)m^{-1}\mathbb{E}\nabla U(Z)$. This is a smooth genuine Hessian and satisfies both equilibrium equations exactly.

Theorem 5.2 (Floor-lowered two-scale frontier; C7–C8)**.** *Uniformly for $0 < \alpha \le \sigma/R$,*

$$N = d+1 \asymp R^2, \qquad \beta \asymp R/\sigma, \qquad \kappa \asymp R/(\alpha\sigma), \tag{5.11}$$

$$L_H \asymp R/\sigma^2, \qquad \tau_H^2 \asymp R/\sigma, \qquad \gamma \asymp \sigma/R. \tag{5.12}$$

Consequently

$$\gamma^{-1} \asymp \frac{R}{\sigma} = N^{1/4}L_H^{1/2}, \qquad (\tau_H^2)^2 \asymp \min\{N, \sqrt{N}L_H\}. \tag{5.13}$$

By lowering α, the same face persists for arbitrarily large κ, including $\kappa > N$.

Proof. The optimal lower curvature is α, because $A(t,y) \to 0$ along a tail. Equation (5.2), the point $y = 0, s = 0$, and $m \asymp R^{-1}$ give the two-sided upper-curvature scale $\beta \asymp R/\sigma$. Lemma 5.1 gives $L_H \asymp R/\sigma^2$.

For the inverse-gap witness take

$$u = e_t, \qquad X = \mathrm{diag}(0, \lambda I_d), \qquad \lambda = \frac{2\delta R}{d}. \tag{5.14}$$

Then $\|X\|_F^2 \lesssim \sigma/R$. Exactly,

$$(2u + XZ)^\top A(2u + XZ) = q(s)\lambda^2\left(\frac{S}{r} - \frac{d}{R}\right)^2 + p(s)\lambda^2\frac{\sigma^2 S}{r^3}. \tag{5.15}$$

The factorization

$$\frac{S}{r} - \frac{d}{R} = (S-d)\frac{\sigma^2 + rR}{rR(r+R)}$$

gives an L^2 bound of order one, while $\mathbb{E}[S/r^3] \lesssim R^{-1}$. Using $q \lesssim \sigma^{-1}$, $m^{-1} \asymp R$, and $\delta^2 \asymp \sigma/R$ in (5.15) yields

$$m^{-1}\mathbb{E}[(2u + XZ)^\top A(2u + XZ)] \lesssim \sigma/R.$$

The floor energy is also $O(\alpha + \sigma/R)$, so the exact Rayleigh identity gives $\gamma \lesssim \sigma/R$ for $\alpha \le \sigma/R$.

For the coupling certificate take $D_0 = \mathrm{diag}(0, I_d/\sqrt{d})$. Compatibility gives

$$e_t^\top T[D_0] = -\frac{(1-\alpha)\delta}{m\sqrt{d}}\mathbb{E}\left[q(s)\frac{S}{r}\right]. \tag{5.16}$$

The central event used in Lemma 5.1 bounds the last expectation below by $c\sqrt{d}/\sigma$. Thus $\tau_H^2 \gtrsim R/\sigma$. The sharp Schur bound gives the reverse inequality

$$\tau_H^2 \le (1-\alpha)(\beta - 1) \lesssim R/\sigma.$$

C1 then gives $\gamma \gtrsim \sigma/R$, completing all two-sided estimates. Finally, $N \asymp R^2$ and $L_H = R/\sigma^2$ imply (5.13). $\square$

Lemma 5.3 (Exact floor interpolation; C8)**.** *If $B \succeq 0$, $\mathbb{E}B = I$, and $W_a = aI + (1-a)B$, then*

$$T_a = (1-a)T_B, \quad H_a = (1-a)H_B, \quad L_{\star,a} = aI + (1-a)L_{\star,B}, \tag{5.17}$$

and therefore

$$\gamma_a = a + (1-a)\gamma_B, \quad \tau_a = (1-a)\tau_B, \quad L_{H,a} = (1-a)L_{H,B}. \tag{5.18}$$

Proof. The constant Hessian has zero first-Hermite and centered-quadratic couplings, so T, H are scaled by $1-a$. Substitute in (2.3); the identity operator commutes with everything, giving the eigenvalue and norm statements. If $\inf_z \lambda_{\min} B(z) = 0$, then as $a \downarrow 0$ the condition number may diverge while $\gamma_a \to \gamma_B$, $\tau_a \to \tau_B$, and $L_{H,a} \to L_{H,B}$. For $a \le \min\{\gamma_B, 1/2\}$ these quantities are comparable to those of B up to numerical factors. □

5.1 Exact-sector numerical triage

Numerics were used only to kill or prioritize candidates, never as a certificate. Hermite quadrature in T and generalized Laguerre quadrature in $S = \|Y\|^2$ reduce every $O(d)$-symmetric expectation to two scalar sectors. The reproducible implementation is `scripts/natural_gradient_local_rate/exact_sector_replication_triage.py`. For J identical modules sharing t, solving the exact equation $J\delta^2 Q = M$ gave:

d	σ	J	δ^2	$J\delta^2$	τ^2_{wit}
16	2	1	0.96532	0.96532	0.24316
16	2	4	0.23228	0.92912	0.25261
16	2	16	0.05751	0.92011	0.25508
81	3	1	0.65248	0.65248	0.64024
81	3	4	0.16106	0.64425	0.64842
81	3	16	0.04014	0.64219	0.65050

The corresponding Rayleigh quotients were $0.688, 0.685, 0.685$ and $0.541, 0.540, 0.539$. The invariant columns suggested the exact isotropy budget proved below; the proof, not this table, is ledger item R2.

6 Rigidity lemmas from failed counterexamples

Each failed counterexample was converted into a proof-usable constraint.

Rigidity lemma 6.1 (Convex mixture and direct sum; R1)**.** *If already-whitened Hessians are mixed as $W = \sum_j \theta_j W_j$, then*

$$\gamma(W) \ge \sum_j \theta_j \gamma(W_j) \ge \min_j \gamma(W_j), \qquad \tau_H(W) \le \max_j \tau_H(W_j), \qquad L_H(W) \le \max_j L_H(W_j). \tag{6.1}$$

If $V(z) = \sum_j V_j(z_j)$ is an orthogonal product, then

$$\gamma(V) = \min\{1, \min_j \gamma(V_j)\}, \qquad \tau_H(V) = \max_j \tau_H(V_j), \qquad L_H(V) = \max_j L_H(V_j). \tag{6.2}$$

Proof. $L_\star$ is affine in W and its smallest eigenvalue is concave, proving (6.1); T and the Lipschitz seminorm are subadditive. For a product, the diagonal covariance sectors reproduce the component operators. Every off-block covariance sector has $T = H = 0$ and eigenvalue one. Gaussian independence gives the stated maximum formulas. Thus rotations, convex randomization, tensor products, and quadratic padding cannot amplify the bad rate. □

Rigidity lemma 6.2 (Shared-longitudinal isotropy budget; R2)**.** *Suppose positive modules A_j share a coordinate t and act on mutually orthogonal transverse blocks E_j, with*

$$\mathbb{E}A_j = \mathrm{diag}(a_j, m_j I_{E_j}).$$

If $0 < \alpha < 1$ and $W = \alpha I + \sum_j c_j A_j$, $c_j > 0$, is exactly isotropic, then

$$\boxed{\sum_j \frac{a_j}{m_j} = 1.} \tag{6.3}$$

For J identical radial-softplus modules, exact isotropy forces

$$J\delta^2 Q = M, \qquad \delta^2 \asymp \frac{\sigma}{JR}. \tag{6.4}$$

The total witness cost, coupling, gap, β, and L_H remain within constants of the one-module values; there is no power of J gain.

Proof. Each transverse mean equation gives $c_j m_j = 1 - \alpha$ and the common t equation gives $\sum_j c_j a_j = 1 - \alpha$. Division proves (6.3). In the identical case this is (6.4). The apparent coherent factor J in the transverse coupling is canceled by $\delta^2 \propto J^{-1}$; the normalized witness has $\|X\|_F^2 \asymp J\delta^2 \asymp \sigma/R$. Summing the exact energy identity (5.15) gives the same scale. The shared longitudinal terms are controlled directly, rather than by a blockwise maximum: in the third-order chain rule, $J\delta^3 = O(1)$, $\sum_j \|h_j\|^3 \leq 1$, and $\delta \sum_j \|h_j\|^2 \leq 1$ for every unit direction $h = (h_t, h_1, \ldots, h_J)$. Likewise, $J\delta^2/\sigma = O(R^{-1})$ controls the shared longitudinal curvature. Together with the blockwise transverse estimates, these give the asserted curvature and Lipschitz scales. □

Rigidity lemma 6.3 (Common-slope radial multi-ramp collapse; R3)**.** *Let $U(t, y) = F(r(y) - R - \delta t)$ with $F', F'' \geq 0$, allowing arbitrarily many ramps. Suppose $W = \alpha I + c\nabla^2 U$, $c > 0$, is exactly isotropic and the nontrivial quantity $Q := \mathbb{E}F''(s)$ is positive. (If $Q = 0$, the isotropizing equation below cannot hold with a nonzero transverse mean.) Let M be the transverse mean in (5.4), and define*

$$\mu_q = \frac{\mathbb{E}[F''(s)S/r]}{Q}, \qquad K = \frac{\sup F''}{Q}.$$

Exact isotropy forces $\delta^2 Q = M$, and for the normalized transverse test

$$|e_t^\top T[D_0]|^2 = \frac{(1-\alpha)^2 \mu_q^2}{\delta^2 d}. \tag{6.5}$$

Moreover

$$K \leq \frac{(\beta - \alpha)\delta^2}{1 - \alpha}, \qquad \mu_q \leq \sqrt{d} + \sqrt{2 \log K}, \qquad \mu_q^2 \leq \delta^2 d \frac{\beta - 1}{1 - \alpha}. \tag{6.6}$$

Thus ramp count produces only one tilted barycenter, not a square-sum gain.

Proof. The isotropy and slice formulas are the same exact differentiations as in Section 5. The peak of F'' is a pointwise Hessian eigenvalue; after applying the isotropic coefficient this gives the first bound in (6.6). The F''-tilted radial density is at most K relative to the Gaussian radial law. Entropy transport plus the one-Lipschitz concentration of $\|Y\|$ gives the second. Finally combine (6.5) with C2. □

Rigidity lemma 6.4 (Positive ridge POVM and conditional energy; R4)**.** *Suppose $a > 0$ and*

$$A(z) = \sum_\ell a_\ell f_\ell(v_\ell^\top z) v_\ell v_\ell^\top, \qquad a_\ell, f_\ell \geq 0, \quad \|v_\ell\| = 1, \quad \mathbb{E}A = aI. \tag{6.7}$$

Omit terms with $m_\ell = 0$. Let $m_\ell = \mathbb{E} f_\ell(G)$, $\mu_\ell = \mathbb{E}[G f_\ell(G)]/m_\ell$, and

$$Q_\ell = \frac{a_\ell m_\ell}{a} v_\ell v_\ell^\top.$$

Then $\sum_\ell Q_\ell = I$, and for every unit w,

$$\left\| \mathbb{E}[(w^\top Z) A(Z)] \right\|_F^2 \le a^2 \sum_\ell \operatorname{Tr}(Q_\ell) \mu_\ell^2 (w^\top v_\ell)^2 \le a^2 \max_\ell \mu_\ell^2. \tag{6.8}$$

If s_ℓ^2 is the variance of G under density $f_\ell(G)/m_\ell$, then for $\eta(z) = 2u + Xz$,

$$\mathbb{E}[\eta^\top A \eta] = \sum_\ell a_\ell m_\ell \left[\|P_{v_\ell^\perp} X v_\ell\|^2 + (v_\ell^\top X v_\ell)^2 s_\ell^2 + \{2 v_\ell^\top u + (v_\ell^\top X v_\ell)\mu_\ell\}^2 \right]. \tag{6.9}$$

Proof. Gaussian regression gives

$$\mathbb{E}[(w^\top Z) A] = a \sum_\ell \mu_\ell (w^\top v_\ell) Q_\ell.$$

The unital positive map $\Phi(c) = \sum_\ell c_\ell Q_\ell$ from the commutative diagonal algebra obeys Kadison's inequality $\Phi(c)^2 \preceq \Phi(c^2)$. Taking traces proves the first inequality in (6.8); the second uses $\sum_\ell \operatorname{Tr}(Q_\ell)(w^\top v_\ell)^2 = w^\top I w = 1$.

For (6.9), condition on $G = v_\ell^\top Z$ and decompose $X v_\ell = (v_\ell^\top X v_\ell) v_\ell + P_{v_\ell^\perp} X v_\ell$. Completing the square under the tilted law gives the three nonnegative terms. Thus a low-gap ridge bank requires simultaneous direction locking, narrow tilted features, and common affine incidence. Uniformly bounded scalar couplings cannot coherently amplify in a fixed positive frame. □

Rigidity lemma 6.5 (Flat quadratic plateau; R5)**.** *If on a transverse region*

$$V(t, y) = a(t) + b(t)^\top y + \tfrac{1}{2} y^\top C(t) y,$$

then $W_{ty} = b'(t) + C'(t) y$. A globally bounded Hessian forces $C'(t) = 0$. If the representation holds only for $\|y\| \le R_0$ and $\|W\|_{\mathrm{op}} \le \beta$, then

$$\|C'(t)\|_{\mathrm{op}} \le \frac{\beta}{R_0} \tag{6.10}$$

up to an inessential factor two. Thus a nonzero flat slice must clip nonlinearly in y and pay an inverse-radius amplitude cost.

Proof. Compare the mixed Hessian block at $y = R_0 v$ and $y = -R_0 v$, then take the supremum over unit v. Letting $R_0 \to \infty$ proves the global statement. Equivalently, full Hessian compatibility reads $\partial_t W_{yy} = \nabla_y W_{ty}$; a constant nonzero transverse slice would integrate to an unbounded mixed block. □

7 Obstruction profiles from stalled proofs

Every proof route that lost a power on the running audit is recorded here as a quantitative construction target.

Obstruction profile 7.1 (O1: Schur-flat high-rank plateau)**.** *Let $B = P_r/\sqrt{r}$. The certified H2 argument gives only*

$$Q_A(B) \le a + L_H \sqrt{r}, \qquad S \le a Q_A(B). \tag{7.1}$$

These data permit $S \asymp L_H \sqrt{r}$, whereas the corrected frontier requires the diffuse excess to be at most $S \lesssim \sqrt{L_H \sqrt{r}}$. A closing proof must use global bounded Hessian clipping and near-Schur saturation jointly, beyond the trace identity. A counterexample target must realize both on the same packet.

Obstruction profile 7.2 (O2: diffuse bounded-density quadratic tilt)**.** *Under (H1), entropy permits a density bounded by κ to tilt the flat chaos $q_{P_r/\sqrt{r}}$ by $\Theta(\sqrt{\log\kappa})$. Nuclear aggregation then permits*

$$S \asymp \sqrt{r}\,\sqrt{\log\kappa}, \tag{7.2}$$

which is exactly the power/log loss in C5. Entropy plus the trace compatibility identity alone cannot prove $S \lesssim \sqrt{r} + \log\kappa$. The missing constraint must connect a diffuse chaos tilt to the mixed Hessian blocks that saturate Schur.

Obstruction profile 7.3 (O3: OU smoothing cannot pay for itself)**.** *For stationary OU smoothing,*

$$\tau_H(P_sW) = e^{-s}\tau_H(W), \qquad L_H(P_sW) \lesssim \frac{(\beta-\alpha)e^{-s}}{\sqrt{1-e^{-2s}}}. \tag{7.3}$$

Undoing the exact e^{-s} contraction removes the apparent smoothing gain and returns at best $1+\sqrt{N}(\beta-\alpha)$. Any successful OU split must estimate a bounded long-time component by structure not inherited from the smoothed Lipschitz modulus alone.

Obstruction profile 7.4 (O4: symmetric tensor without global clipping)**.** *The abstract symmetric tensor*

$$G = c\,\mathrm{Sym}(e_1 \otimes P_r) \tag{7.4}$$

has $\|G\|_{\mathrm{inj}} \asymp c$ but $\|G(e_1,\cdot,\cdot)\|_F^2 \asymp rc^2$. Therefore symmetry and the a.e. injective bound in (H2) alone cannot produce the desired exponent. The missing input is that G is the averaged third derivative of a Hessian confined globally between two scalar barriers. Rigidity R5 is the simplest manifestation of this clipping cost.

Obstruction profile 7.5 (O5: surviving counterexample geometry)**.** *R1–R5 rule out convex mixtures, disjoint products, one-level shared-t stars, common-slope multi-shells, fixed positive ridge frames, and literal flat quadratic plateaux as amplification mechanisms. A surviving adverse family must therefore use at least one of:*

(a) variable-slope overlapping radial packets satisfying exact isotropy $\sum_j \delta_j^2 Q_j = \sum_j M_j$ and sharing one affine witness;
(b) a genuinely overlapping hierarchy that reuses transverse mean beyond the nodewise budget (6.3);
(c) signed component Hessians whose total remains pointwise positive, with deterministic curvature and Lipschitz control;
(d) a near-incidence slab packing satisfying all three nonnegative conditions in (6.9) while preventing pointwise curvature stacking.

In every case the mixed coupling must nearly saturate the shifted Schur bound on the same high-nuclear-rank packet. This is a construction target, not a belief that such a family exists.

8 Mandatory running audit on $(d,\sigma) = (n^4, n)$

The exact two-scale certificate gives

$$N \asymp n^4, \quad \alpha \asymp n^{-1}, \quad \beta \asymp n, \quad \kappa \asymp n^2, \quad L_H \asymp 1, \quad \tau_H^2 \asymp n, \quad \gamma^{-1} \asymp n. \tag{8.1}$$

The audit is:

Certified step	Value on family	Verdict
Shifted Schur ab	$\asymp n$	sharp
H2 rank branch $\sqrt{N}L_H$	$\asymp n^2$	power-slack by n; O1
Direct Stein NL_H^2	$\asymp n^4$	power-slack by n^3
H1 entropy branch	$\asymp n^2\sqrt{\log n}$	power-slack by $n\sqrt{\log n}$; O2
OU bridge	at least $\asymp n^3$ in the tested split	power-slack by n^2; O3
Corrected excess conjecture	$\min\{N, \sqrt{N}L_H\} = n^2 = (\tau_H^2)^2$	exactly saturated

The geometry responsible for every loss is explicit. The averaged t-slice has $d \asymp n^4$ coherent transverse eigenvalues of order $n^{-3/2}$: its operator norm is tiny, while its Frobenius square is n. It is generated by a shell with tangential curvature of order one and longitudinal width of order $n^{3/2}$. Any proof that collapses the slice to a trace, nuclear norm, or operator norm loses a power here and is declared non-closing rather than recycled with new constants.

For the floor-lowered family, write $r = R/\sigma$. Then

$$\sqrt{N} = R, \qquad L_H = r^2/R, \qquad \tau_H^2 \asymp r, \qquad \sqrt{\kappa} \geq r,$$

and

$$\min\{N, \sqrt{N}L_H\} = r^2 \asymp (\tau_H^2)^2. \tag{8.2}$$

Thus lowering the floor does not expose a missing κ dependence in the corrected target; it exactly saturates its regularity/dimension side.

9 The standalone closing lemma

The proof and refute ledgers meet at one precise statement. It is not proved in this report.

Open problem 9.1 (Standalone excess-coupling lemma, H2 version)**.** *There should be a numerical constant $C > 0$ such that, for every admissible genuine Hessian and every unit w,*

$$\boxed{(\|M_w\|_F^2 - Cd_{\kappa,L_H})_+^2 \leq C\min\{N, \sqrt{N}L_H\}, \qquad d_{\kappa,L_H} = 1 + \min\{\log\kappa, \log(e + L_H)\}.} \tag{9.1}$$

Taking square roots would give

$$\tau_H^2 \leq C\left[d_{\kappa,L_H} + \min\{\sqrt{N}, N^{1/4}L_H^{1/2}\}\right].$$

Combining this with C1–C2 would prove the upper side of the corrected conjecture (1.6). It has the right scalar reduction; it is saturated by the unit, two-scale, and floor-lowered radial families; and every audited inequality that currently fails to prove it is isolated in O1–O4.

Under (H1) alone, the corresponding standalone target is

$$\boxed{(\|M_w\|_F^2 - C(1 + \log\kappa))_+^2 \leq CN.} \tag{9.2}$$

It would imply $\tau_H^2 \lesssim 1 + \log\kappa + \sqrt{N}$ and, after the curvature-floor case split, the upper side of (1.5).

The sharp unresolved issue is therefore not a symmetry-only first-Hermite estimate. It is a *diffuse-excess theorem*: after removing the scalar entropy/Lipschitz contribution, prove that a high-rank first-Hermite Hessian slice cannot simultaneously have large Frobenius energy and near-saturate the shifted Bochner–Schur inequality without paying the global bounded Hessian clipping cost. This formulation uses full Hessian compatibility, places (H2) exactly in the regularity branch, and survives every certified family and rigidity audit in this run.

10 Final bracket

The guaranteed deliverable is therefore the following nonconditional bracket.

$$\begin{aligned}
\text{(H1) lower:}\quad & \mathfrak{R}_D(N,\kappa) \gtrsim 1+\log\kappa+\sqrt{\min\{N,\kappa\}}, \quad \text{(supplied certified families)},\\
\text{(H1) upper:}\quad & \mathfrak{R}_D(N,\kappa) \lesssim 1+\min\{\sqrt{\kappa}, \sqrt{N\log(e\kappa)}+N^{1/3}\log(e\kappa)\},\\
\text{(H2) adverse face:}\quad & \mathfrak{R}_L(N,\kappa,L) \gtrsim 1+\min\{\log\kappa,\log(e+L)\}+\min\{\sqrt{\kappa},\sqrt{N},N^{1/4}L^{1/2}\},\\
& \qquad\qquad \text{on the parameter faces certified above,}\\
\text{(H2) upper:}\quad & \mathfrak{R}_L(N,\kappa,L) \lesssim \text{the right side of (3.23), with } L_H \text{ replaced by } L.
\end{aligned}$$

The original F–L statement is rejected by Theorem 1.1. The corrected F–L and the original F–D statements remain open. No genuine-Hessian, exactly whitened family was found above the corrected frontier; no closing proof was found either. Every asserted inequality and family in this report has an explicit proof or certificate, while the boxed statements (9.1) and (9.2) are visibly marked as the remaining open lemmas.